Aashray Jain

Software Developer | Full-Stack & AI/ML Engineer

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Summary

Software Engineer and Full-Stack & AI/ML Specialist with 4+ years of professional engineering experience across enterprise backend systems, scalable cloud architectures, and production LLM orchestration. Key contributor at Novafy OU rebuilding the NovaAgent v2 engine in Python/FastAPI with Model Context Protocol (MCP) tool interfaces, prompt-caching (∼99% cached input tokens), and ERP Human-in-the-Loop validation. Former TCS Systems Engineer with a proven track record delivering BFSI real-time risk platforms (500K+ users) and reducing cloud infrastructure costs by 70%. Inventor of ArihantQNet (novel tri-branch XAI diagnostic architecture) and officially registered copyright holder for ArihantQInvoice ERP. Currently pursuing M.Tech. in AI & Machine Learning at NIMS University Rajasthan.

Education

- **M.Tech. in Artificial Intelligence & Machine Learning** *2026 – 2028*
  NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan Pursuing
- **B.Tech. in Electronics & Communication Engineering** *2017 – 2021*
  Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh (RGPV Bhopal) 8.74 CGPA
- **Senior Secondary (81.4%) & Secondary (8.0 CGPA)** *2015 – 2017*
  R.S. Memorial H.S. School (MP Board) & DAV ACC Public School (CBSE Board), Katni, MP

Experience

- **Novafy OU** *Oct 2026 – Present*
  Software Development Intern Remote (Hamburg, Germany)
  - Core contributor to the ground-up rebuild of **NovaAgent v2 conversational engine** in Python, FastAPI, and PostgreSQL, authoring **190+ commits** across a reviewed pull-request engineering workflow.
  - Architected a plug-in LLM chat engine consuming and serving tools over **Model Context Protocol (MCP)**, featuring per-tenant tool configurations, multi-model fallback chains, and dynamic database-backed model catalogs.
  - Engineered a **Human-in-the-Loop (HITL) safety layer** for Microsoft Dynamics 365 Business Central ERP; built pre-execution metadata validation, one-click batch approvals, and value-substitution preventing hallucinated lookup codes.
  - Optimized latency and operational costs via provider-side **prompt-caching achieving ∼99% cached input tokens**; developed conversation compaction and fit-to-window algorithms preventing context overflow.
  - Built an automated evaluation harness with 70+ conversational test scenarios to benchmark prompts and models; wrote comprehensive integration suites running against isolated throwaway PostgreSQL test containers.
  - Hardened security with tenant and session isolation, replay protection, idempotent MS Teams redelivery handling, structured telemetry, and active outage metering/alerting.
- **ArihantQ** *Aug 2025 – Present*
  Founder (Part-Time and Experimental Venture) Remote (India)
  - Established a personal AI innovation lab focused on Generative AI, RAG, and LLM applications to build production prototypes and strategically advance technical capabilities.
  - Architected ArihantQInvoice, a full-stack ERP platform for billing, inventory, GST invoicing, stock tracking, and supplier operations using Node.js, Express.js, SQLite, and Sequelize; officially registered software copyright with Copyright Office, Govt. of India.
  - Validated concepts in national hackathons: Lyzr AI Architect Challenge prize winner (AI Cost Optimization Advisor), Elastic AI Bengaluru attendee, and public recognition from MariaDB CEO and Foundation.
  - Secured full business enablers (Udyam Registration, Shop and Establishment Act License, and Current Bank Account) providing a legal operational foundation for independent technology ventures.
- **Tata Consultancy Services** *Jul 2021 – Jan 2025*
  Systems Engineer Mumbai & Kochi, India
  - Developed secure real-time alert analysis systems and Line of Business (LOB) dashboards for a Canadian BFSI client using React, Node.js Lambda, and DynamoDB, improving risk monitoring efficiency by 90%.
  - Built and maintained scalable REST APIs using Node.js and Express.js supporting secure financial workflows and compliance-driven enterprise applications.
  - Led a multi-account AWS cloud migration for a large-scale platform, optimizing infrastructure architecture and reducing operational costs by 70% while improving resilience.
  - Strengthened enterprise security posture by integrating SAML authentication and AWS Cognito across applications serving 500K+ users, reducing vulnerabilities by 40%.
  - Automated business reporting workflows using Angular, NX Monorepo, Python (Pandas, Flask), and Excel automation, reducing manual effort by 15+ hours/week.
  - Collaborated with TCS Rapid Labs on AI/ML PoCs across NLP, Computer Vision, and Generative AI; previously developed Asset Procurement module for EquipMax 3.0 during internship (Angular/Ionic).

Projects

- **Class-Balanced Benchmarking & XAI Clinical DSS (ArihantQNet)** *Oct 2026*
  PyTorch, TorchVision, Python, Streamlit, Grad-CAM (XAI), Scikit-learn, OpenCV Live App | GitHub
  - Architected an end-to-end deep learning and Explainable AI (XAI) clinical intelligence framework for multi-stage Alzheimer's Disease stratification from 6,400 T1-weighted axial brain MRI scans (70/15/15 stratified partition).
  - Engineered Normalized Square-Root Inverse-Frequency Weighted Cross-Entropy Loss with AdamW, compressing 49.78:1 class skew to 7.06:1 and achieving 100% sensitivity on rare Moderate Demented cases.
  - Formulated novel tri-branch hybrid architecture **ArihantQNet**, fusing ResNet50 (macro-contextual key), DenseNet121 (fine-boundary query), and EfficientNet-B0 (channel-calibrated value) via 1×1 bottleneck projections, Quantum-Inspired Multi-Head Cross-Attention (Q-MCA), and SE recalibration.
  - Integrated Grad-CAM neuroanatomical heatmaps (hippocampus and ventricular atrophy) and deployed an interactive 6-module Streamlit clinical intelligence workbench for live inference and diagnostic reporting.
- **ArihantQInvoice Billing & Inventory Management System** *Apr 2026*
  Node.js, Express.js, SQLite, Sequelize, EJS, Swagger, DOCX Generation
  - Designed and deployed a production-ready ERP platform digitizing retail operations; implemented GST/non-GST billing, stock tracking, and automated DOCX invoice generation across 50+ REST APIs with OpenAPI/Swagger docs.
  - Officially registered software copyright for ArihantQInvoice Billing and Inventory Management with the Copyright Office, Government of India.
- **AI-Powered Portfolio & Public Profile WebApp** *Jul 2025*
  HTML, CSS, JS, Bootstrap 5, GitHub Pages Project URL | GitHub
  - Developed responsive client-side portfolio showcasing full-stack, AI, and Quantum projects; integrated LinkedIn public badge sync, SEO metadata, dark/light theme, and CI/CD deployment via GitHub Pages.
- **RAG-Powered PDF Question Answering App** *Jul 2025*
  Python, Streamlit, LangChain, HuggingFace, FAISS Live App | Hugging Face
  - Engineered document Q&A application using RAG, enabling users to upload PDFs and query contents in plain English; powered by FAISS vector index and multi-model flexibility via HuggingFace (DeepSeek, Mixtral, Zephyr).
- **AI Cost Optimization Advisor (Lyzr AI Architect Challenge)** *Jun 2025*
  Lyzr Studio, AWS Bedrock, Claude, GPT, Gemini, Mistral Live Agent | GitHub
  - Built GenAI agent analyzing AI use cases and recommending cost-optimal models with ROI estimation; indexed 25+ knowledge sources for RAG, simulating business workflows and achieving documented 666% ROI.

Technical Skills

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL
- **AI / ML & GenAI Frameworks:** FastAPI, Model Context Protocol (MCP), PyTorch, LangChain, Hugging Face, FAISS, Prompt-Caching, Streamlit, Grad-CAM (XAI)
- **Backend Architecture:** Node.js, Express.js, REST APIs, Microservices, PostgreSQL, SQLite, DynamoDB, Human-in-the-Loop (HITL) Workflows
- **Frontend Development:** ReactJS, Angular, HTML5, CSS3, Bootstrap 5, Material UI, D3.js
- **Cloud & Enterprise DevOps:** AWS (Lambda, API Gateway, S3, Cognito, DynamoDB, CDK), GCP (Vertex AI), Docker, Git, CI/CD, Swagger/OpenAPI
- **Security & Ecosystems:** SAML, AWS Cognito, CyberArk, RBAC | Microsoft 365, MS Teams App, Dynamics 365 Business Central, SharePoint, Outlook

Achievements

- **Top 100 Winner** — Microsoft AI Odyssey Challenge by Microsoft & TCS; 1920-min Azure AI-102T00 program. *2024*
- **Microsoft Learn Honors** — Earned 49 Badges, 10 Trophies, 48 Module completions, and 2 Applied Skills Certifications. *2024*
- **National AI Recognition** — **FutureSkills Prime (AI Applications)** by NASSCOM & Govt. of India; MyBharat Honors. *2025*

Certifications

- **Cloud & AI Systems:** **Microsoft**: Applied Skills: GenAI with Azure OpenAI | **Microsoft**: Applied Skills: Document Processing with Azure AI | **AWS**: Cloud Practitioner Essentials (Second Edition)
- **Google Cloud Badges:** GenAI Apps with Gemini & Streamlit | Explore GenAI with Vertex AI Gemini API | Gemini Multimodality & Multimodal RAG | Text Prompt Engineering Techniques
- **Data Science & Foundations:** **IBM**: Applied Data Science with Python (Level 2) | **IBM**: Big Data Foundations (Level 2) | **IBM**: Artificial Intelligence Fundamentals
- **Academic & Developer Credentials:** **IIT Kanpur (NPTEL)**: Programming in C | **IIT Kharagpur (NPTEL)**: Programming in Java | **HackerRank**: Java (Basic) & Gold Badge | **SEBI** Investor Certification (ID: NISM20250000203277)

Extracurricular & Volunteering

- Active participant in GenAI hackathons, AI webinars, and competitive developer communities; TCS Blood Donor (Amrita Hospital, Kochi); participant in national technical fests (Planet Engineers, Convolution) and TEDxLNCTWomen.